Magnitudes: How Bright Is It Really?

Covers Lecture 3.1 about 75 minutes

Name: _____ Date: _____ Marks: _____ / 70

You may use: a calculator (with log/power keys) **or** the table below.

Δm	0.5	1	1.5	2	2.5	3	4	5	10	15
ratio	1.6	2.5	4.0	6.3	10	16	40	100	10^4	10^6

Formula box

$$\frac{F_1}{F_2} = 10^{0.4(m_2-m_1)} \qquad m_1 - m_2 = -2.5\log_{10}\frac{F_1}{F_2} \qquad M = m - 5\log_{10}\frac{d}{10\text{ pc}}$$

$$d = 10^{(m-M+5)/5}\text{ pc} \qquad \frac{L}{L_\odot} = 10^{0.4(4.83-M)} \qquad 1\text{ pc} = 3.26\text{ ly}$$

Section A — Fill in the blanks ★ 8 marks

1. On the magnitude scale, a **smaller** number means a _____ object.
2. A difference of 5 magnitudes means a brightness ratio of exactly _____.
3. A difference of 1 magnitude means a brightness ratio of _____.
4. The modern magnitude scale was defined by _____ in 1856; he later directed the _____ Observatory in India.
5. The original 6-class magnitude system was invented by the Greek astronomer _____.
6. Absolute magnitude is the apparent magnitude a star would have at a distance of _____ parsecs.
7. 1 parsec = _____ light-years.
8. The absolute magnitude of the Sun is about _____.

Section B — Multiple choice ★ 8 marks

1. Which is the brightest? (a) $m = 2$ (b) $m = 0$ (c) $m = -1$ (d) $m = 5$
2. A magnitude 1 star compared with a magnitude 6 star is: (a) 5× brighter (b) 100× brighter (c) 100× fainter (d) 6× brighter
3. A lamp moved to twice the distance looks: (a) 2× fainter (b) 4× fainter (c) 8× fainter (d) unchanged
4. Deneb ($m = 1.25$) looks fainter than Sirius ($m = -1.46$) because Deneb: (a) is less luminous (b) is much farther away (c) is cooler (d) is a planet
5. The faintest naked-eye stars from a dark site are about: (a) $m = 2$ (b) $m = 4$ (c) $m = 6.5$ (d) $m = 10$
6. A distance modulus $m - M = 0$ means the star is at: (a) 1 pc (b) 10 pc (c) 100 pc (d) 0 pc

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7. The colour index $B - V$ of a red star is: (a) large and positive (b) zero (c) negative (d) always -1
8. Algol changes brightness because: (a) it pulsates (b) a companion star eclipses it (c) clouds pass in front (d) it is a planet

Section C — Brightness ladder ★ 8 marks

C1 (4). Arrange from **brightest to faintest**: Polaris, Full Moon, Sirius, Sun, Venus (at its brightest), naked-eye limit, Jupiter, Vega.

_____ → _____ → _____ → _____ → _____ → _____ → _____ → _____

C2 (4). Match each object to its magnitude: -26.7, -12.7, -4.6, -1.5, 0.0, 2.0, 6.5, 14.4

Object	m	Object	m
Pluto		Sun	
Sirius		Vega	
Full Moon		Venus (max)	
Polaris		faintest naked-eye star	

Section D — Calculations ★★ 36 marks

Worked example

How many times brighter does Arcturus ($m = -0.05$) look than Spica ($m = 0.97$)? $\Delta m = 0.97 - (-0.05) = 1.02$; ratio $= 10^{0.4 \times 1.02} = 10^{0.408} \approx$ **2.6**.

D1. [6] How many times brighter is (a) a magnitude 1 star than a magnitude 4 star? (b) a magnitude 0 star than a magnitude 10 star? (c) Venus (-4.6) than Sirius (-1.5)?

D2. [4] (a) A star is **40 times fainter** than Vega ($m = 0$). What is its magnitude? (b) Star A is **10 times brighter** than star B ($m = 6.0$). What is A's magnitude?

D3. [4] (a) A star is moved **10 times farther away**. By how many magnitudes does it get fainter? (b) And if moved **3 times farther**? (Hint: $2.5 \log_{10} 9 = ?$)

D4. [6] Find M: (a) $m = 6.0$, $d = 100$ pc (b) $m = 2.0$, $d = 10$ pc (c) $m = 10.0$, $d = 1{,}000$ pc.

D5. [6] (a) A star has $m = 8$ and $M = 3$. Find its distance in parsecs and light-years. (b) A supergiant with $M = -3$ appears at $m = 12$. How far is it? Is it inside our galaxy ($\sim 30{,}000$ pc across)?

D6. [3] Alpha Centauri is 1.34 pc away. If our Sun ($M = 4.83$) were there, what would its apparent magnitude be? Which bright star would it rival?

D7. [3] Arcturus has $M = -0.3$. How many times more luminous (visually) is it than the Sun?

D8. [2] Algol varies between $m = 2.1$ and 3.4. By what factor does its brightness change?

D9. [2] A star of $m = 1.0$ overhead appears as $m = 2.0$ near the horizon in Delhi's winter haze. What fraction of its light is lost?

Section E — Practical: estimate magnitudes by eye ★★ 10 marks

E1 (5). On a clear evening when the Big Dipper (March–June) or Orion (December–March) is up, **rank its 7 stars** from brightest to faintest by eye, without looking at a table. Then compare with the catalogue values:
Big Dipper: Dubhe 1.79, Merak 2.37, Phecda 2.44, Megrez 3.31, Alioth 1.77, Mizar 2.23, Alkaid 1.86.
Orion: Rigel 0.13, Betelgeuse ∼0.5 (variable!), Bellatrix 1.64, Alnilam 1.69, Alnitak 1.77, Saiph 2.09, Mintaka 2.23.
Which star did you misjudge most, and why (colour? height above the horizon? nearby lights?)

E2 (5). Using **Argelander's step method** with two comparison stars, estimate the magnitude of **Megrez** or **Saiph**. Show your comparison stars, their magnitudes, your step estimates and the final value.

Bonus ★★★ up to 6 marks

1. (3) Two stars, **each** of magnitude 3.0, are so close that they look like one point. What is the magnitude of the combined point?
2. (3) The Sun is 4×10^5 times brighter than the Full Moon. Express this as a magnitude difference and check it against -26.7 and -12.7.